

Faceless Studio — Control & Quality Systems

A navigation guide: every quality gate, control, and re-render system — with your human-in-the-loop buttons and the machine's automated functions. One asset (a video) is a single row that advances through the checkpoints below; you steer it from the **Library tile** (quick actions) or the **Asset Canvas** (full edit).

HUMAN

Human-in-the-loop button

A control you press. Nothing behind it publishes or spends without this unless autonomy is raised.

AUTO

Automated function

Runs by itself (cron / agent / pipeline).
Produces a result you review.

GUARD

Guardrail / control

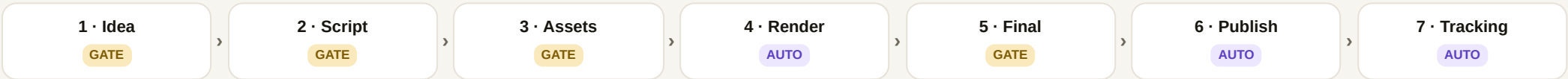
A hard stop: budget, kill switch, revision cap, fail-closed.

LEARN

Learning loop

Feeds outcomes back so the system improves over time.

The asset lifecycle — 7 checkpoints, 4 review gates



Amber = a review gate the pipeline pauses at (in Assist). **Co-pilot** auto-approves high-confidence gates; **Autopilot** resolves them all — but a low quality score or a policy risk always holds for you.

STAGE	AUTOMATED QUALITY FUNCTIONS (RUN BEFORE YOU LOOK)	YOUR BUTTONS & WHERE	GUARDRAILS
1 · Idea	Idea scored against a floor + one improvement round; lexical & semantic (pgvector) near-dup rejection; differentiation / demand-evidence check; performance-weighted ranking after cold-start.	Approve Changes Kill on the tile · Queue / Dismiss for intelligence idea cards · Generate ideas . Library → Ideas section, or the Canvas Idea checkpoint.	Idea-gate score floor blocks weak ideas before any spend.
2 · Script	QC review (0–10) + revise-once; editorial guard (legal/spam/metadata); search-grounded fact-check (holds above a risk cap); QC-lessons fed into the writer.	Approve & continue Request changes Kill · edit a beat + re-voice · on the QC verdict. Canvas → Script checkpoint + Script editor.	Fact-check risk ≥ cap → hold. Grader-down → hold (never "publish anyway").
3 · Assets	Budget reserved before the batch; FLUX prompt pre-check; seed-still vision gate; blank-frame pixel check; pHash variety re-roll; per-beat relevance-to-narration vision score + auto re-roll.	Approve → render Changes Kill · crown a thumbnail · reroll a clip (+ steer note) · Source Library (licensed footage). Canvas → Assets checkpoint.	Fail-closed: paid provider live but QC model mocked → block spend. Per-video & monthly budget caps re-checked before rerolls.
4 · Render	Render farm (GitHub Actions) assembles the MP4 + Short; ffmpeg media-QC (black/frozen frames, silence, LUFS loudness); data-viz chart verification; bounded render-retry (×3) then surfaced.	Resume / Reset & retry if a render failed. Library tile / Canvas header. Otherwise passive — the farm advances it.	Hard media-QC fail holds structurally-broken cuts. Retry capped at 3, then held for you.
5 · Final	Frame critic (vision); Self-Watch gate — timing, script → visual match, competitive fit, transitions; auto-fix loop (critique → fix → re-render, bounded); final QC → posting-privacy mapping; editorial re-check.	Approve final cut Request changes Kill · on the Self-Watch verdict · step-back a stage. Canvas → Final checkpoint + Publish Kit.	Self-Watch overall < floor or a content-policy risk → hold for a human even on Autopilot. Revision hard-cap (4).

6 · Publish	Scheduler releases approved auto-pilot videos at their slot; render farm uploads to YouTube at the QC-decided privacy (public/unlisted); attribution + AI-disclosure injected.	Publish to YouTube / Mark as uploaded · download MP4/Short/thumb · copy title/description/tags. Canvas → Publish Kit.	Publish is a one-way latch (no double-post). Manual until a channel clears YouTube monetization review (recommended).
7 · Tracking	Nightly stats ingestion (views/retention/engaged-views); revenue estimate; feeds the learning loops (below). Published assets never show as "awaiting you."	Refresh stats · Derive Shorts from a long-form. Canvas → Tracking panel · Library → Published section.	Live analytics failures skip (never fabricate data into the ledger / learning loop).

Faceless Studio · every mutation flows through the same server actions whether you click a Library tile, the Asset Canvas, Telegram, or the studio-MCP — so the gates, budgets, and kill switch apply identically everywhere.

Control systems — your hard stops

Kill switch GLOBAL

- **Human:** toggle in **Settings** → **Pipeline control**. On → every pipeline run halts and holds with a visible reason; failure to set surfaces an error (a safety control never fails silently).
- Checked at the top of every run, every operator tick, and every paid clip generation.

Budget caps SPEND

- **Auto:** one ledger + `checkBudget` enforces **per-video**, **monthly**, and the operator's **30-day cycle** cap. A paid stage past a cap → held with the reason.
- **Human:** set caps in the project wizard / Settings; raise a cap then Resume the held tile.
- Provider circuit-breaker: `fal balance-exhausted` → pauses dependent work + notifies, auto-clears on a healthy check.

Revision cap LOOP BOUND

- **Auto:** warns at revision #3, **hard-blocks at #4** (per video). Autopilot's own single revision is exempt.
- Prevents an asset from burning spend in an endless "Changes" loop — approve, kill, or step back instead.

Autonomy tiers PER GATE

- **Human:** set **Assist** / **Co-pilot** / **Autopilot** per gate (project wizard & Settings) and the operator mode in **Autopilot**.
- **Assist** = always ask. **Co-pilot** = auto-approve only high-confidence (QC ≥ 7.5). **Autopilot** = resolve all gates, but low-score / policy-risk still holds.

Re-render & fix ladder — how a weak render gets better

At **Final review** a below-threshold render climbs this ladder automatically (bounded by `maxRenders` + a spend cap), and you can act at any rung.

RUNG	WHAT HAPPENS	KIND	BOUND / CONTROL
Self-Watch re-rolls	The pre-publish gate flags beats whose visual doesn't match the narration; those feed the auto-fix loop as targeted re-rolls (one integrated fixer).	 AUTO	Only beats not already flagged; bounded by the render cap.
Auto-fix (Layer 1)	Vision critique → up to 2 coherent visual fixes (reframe, tighten captions, re-choreograph, licensed-footage swap) → re-render → re-critique. Score-banded model.	 AUTO 	<code>maxRenders</code> + <code>spendCap</code> ; a low render that doesn't respond is held immediately. Fixes that move QC ≥0.3 become governed playbook lessons.
Layer 2 — guidance hold	When no <i>visual</i> fix applies but the failure is structural (starved beat, weak hook, pacing), a banded model writes a specific revision brief and holds it as the pause reason — one-tap actionable.	 AUTO →  HUMAN	Always the safe fallback. You read the brief and press Request changes / edit the script.
Auto-Rescript	<i>Optional, off by default.</i> Instead of holding, it re-writes the script from that brief and re-flows the video to a fresh render — once per video , then holds. Outcome-guarded (a non-improving rescript is held).	 AUTO  GATED	Env kill-switch <code>AUTOFIX_AUTO_RESCRIPT</code> = <code>off</code> (default) / <code>dry</code> (log only) / <code>on</code> ; auto-pilot videos only; global kill switch honored; once per video.
Manual fixes	Reroll a beat with a steer note, swap in licensed footage, re-pick the thumbnail, edit a script beat & re-voice, or step back a whole stage.	 HUMAN	Revision cap applies to "Changes"; step-back can optionally discard assets (VO is cached either way).

Learning loops — why it improves over time

🧠 Judge calibration (C1)

- **Human:** 👍 / 👎 on QC & Self-Watch verdicts (Canvas checkpoints). ~50 labels make each rubric criterion measurable (agreement %).
- Surfaced in **Settings** → **judge calibration**.

🧠 Outcome audit (C3)

- **Auto:** nightly join of gate scores vs day-7/28 retention. If a gate's score stops predicting retention, the judge is drifting/gamed — flagged for a rubric rewrite.

🧠 Governed memory (C4/C8)

- **Auto:** evidence-gated "Do/Avoid" lessons per craft namespace, decay + re-confirm, nightly librarian curation. Only lessons backed by a real QC/retention delta persist.

🧠 Optimizer & bandit

- **Auto:** weekly insights; template proposals auto-apply as a **3-video canary** (auto-revert if they underperform). Format bandit tilts the content mix once a channel clears ~8 published videos.
- **Human:** **Apply** / **Dismiss** insights in the **Feed**.

Quick answers to two things you asked:

- **"Resume" on a "Ready to publish" item:** it only appears if that approved video is *paused* (e.g. a budget cap was hit or the kill switch was on). It re-runs the pipeline to clear the pause — it does **not** publish. To actually publish, open the tile → **Publish Kit** → **Publish to YouTube / Mark as uploaded**. (A clean approved video shows no Resume — just tap it to open the Publish Kit.)
- **"Changes" on Ideas / Script / Production items:** it sends the asset back for a revision *with your note*. Idea → sharpens the idea in place; Script → rewrites the script from your note; Assets/Production → regenerates the assets. It counts against the revision cap (warn #3, hard-stop #4).